

Graph-Aware Anomaly Detection for Blockchain Forensics: A 48-Experiment Benchmark on the Elliptic Bitcoin Dataset

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Abstract—Detecting illicit activity on a public blockchain involves facing three concurrent challenges: severe class imbalance (2 % illicit among all transactions), temporal concept drift (illicit rate shifting from 11.6 % in training to 6.5 % at test time), and a pseudonymous graph whose nodes carry 165 anonymous features. We present a systematic benchmark of 48 model variants across five paradigms—unsupervised, supervised, deep neural, graph neural network (GNN), and semi-supervised—on the Elliptic Bitcoin dataset (203 769 transactions, 234 355 directed edges), organized into five iterative improvement rounds. All models share a strict temporal train/test split that respects the time-series nature of the data. The best result, $F1 = 0.8265$, is achieved by Gradient Boosting augmented with seven structural topology features. GraphSAGEv2 reaches $F1 = 0.698$ using graph message passing over node features and topology, recovering 84.5 % of the supervised ceiling. Classical feature-space unsupervised baselines never exceed $F1 = 0.076$; the course-derived DOMINANT graph autoencoder reaches $F1 = 0.212$. A controlled GNN ablation shows that LayerNorm and self-loops (+0.068 $F1$) matter more than depth or attention. β -VAE with $\beta=0.01$ achieves $F1 = 0.407$, the best purely unsupervised result.

Index Terms—anomaly detection, blockchain forensics, GraphSAGE, DOMINANT, gradient boosting, β -VAE, EVT threshold, SHAP, Optuna, class imbalance, Elliptic dataset

I. INTRODUCTION

Bitcoin’s blockchain is fully public yet pseudonymous: every transaction is permanently recorded, but the identities behind wallet addresses remain hidden. This pseudonymity has made Bitcoin a vehicle for money laundering, ransomware payments, and darknet market settlements. Traditional anti-money-laundering (AML) controls rely on verified identity, while blockchain forensics must infer intent from transaction patterns and graph structure.

The Elliptic dataset [1] is the largest publicly available labeled graph of real Bitcoin transactions: 203 769 transactions, 234 355 directed edges, 4 545 labeled illicit (2 %), 42 019 labeled licit (21 %), and 157 205 unlabeled (77 %) over 49 time steps spanning roughly two years. Each node carries 165 anonymous features, 93 local statistics and 71 aggregated neighborhood summaries (some sources count 166 including the time-step index, which we treat as a split attribute rather than a predictive feature).

Three intertwined challenges:

- 1) **Severe imbalance.** 9.76 % illicit among labeled nodes; 6.5 % in the test window. A naive majority classifier achieves 93.5 % accuracy while detecting zero illicit transactions, making accuracy a completely uninformative metric.
- 2) **Temporal concept drift.** Illicit rate drops from 11.6 % to 6.5 % from train to test (44 % relative decline). Random shuffling leaks future patterns into training, inflating every metric.
- 3) **Anonymous features.** Elliptic has not published feature semantics, so cross-model feature agreement is the only validation of signal robustness.

A. Research Questions

- **RQ1:** Do classical unsupervised detectors provide a viable AML baseline, or do illicit transactions fail to form anomalous clusters?
- **RQ2:** How much does graph structure add over tabular supervised models, and what fraction of the supervised ceiling can a GNN recover?
- **RQ3:** Which GNN design choices (depth, width, normalization, attention) matter most on a sparse graph (avg. degree 2.3)?
- **RQ4:** Do any features rank consistently across heterogeneous model families?

B. Contributions

- 1) Strict **temporal evaluation protocol** (steps 1–34 train / 30–34 validation / 35–49 test) applied to all 48 variants.
- 2) **48-variant benchmark** across five improvement rounds covering classical anomaly detection, supervised trees, deep autoencoders, GNNs, semi-supervised pseudo-labels, and topology-augmented methods.
- 3) **10-experiment GNN ablation** isolating hidden dimension, depth, LayerNorm, self-loops, and attention heads.
- 4) **Optuna TPE tuning** [14] (40 trials each) for GBM, RF, and GraphSAGE.

- 5) Evaluation of **course-derived methods**: DOMINANT [6], LSTM-AE, β -VAE [7], spectral embedding, EVT thresholding [15], and pseudo-label propagation [16].
- 6) **Cross-paradigm explainability** via SHAP and gradient attribution.

II. RELATED WORK

Weber et al. [1] introduced the Elliptic dataset and benchmarked LR, RF, and GCN, but did not explore unsupervised methods, deeper GNN architectures, or the score-inversion phenomenon in reconstruction-based detectors.

GNNs for fraud. Hamilton et al. [3] introduced GraphSAGE; Velickovic et al. [4] proposed GAT; Brody et al. [5] fixed GAT’s expressivity with GATv2. Pareja et al. [2] extended GCN to dynamic graphs (EvolveGCN).

Graph anomaly detection. Ding et al. [6] proposed DOMINANT, a GCN autoencoder that jointly reconstructs node attributes and graph structure without labels.

Reconstruction-based detection. Standard autoencoders [8] flag high reconstruction error as anomalous. On Elliptic we observe inversion: illicit transactions reconstruct with *lower* MSE [9]. Denoising AE [10] and VAE [11] are evaluated.

β -VAE [7] adds weight β to the KL term in the ELBO:

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - \beta D_{\text{KL}}(q_\phi(z|x) \| p(z)). \quad (1)$$

Lower β is reconstruction-dominated; $\beta=0.01$ yields best F1 (0.407) since reconstruction dominance preserves the inversion signal.

EVT thresholding. The Generalized Pareto Distribution (GPD) models score tails to set thresholds without labels [15]:

$$t_{\text{EVT}} = u + \frac{\hat{\sigma}}{\hat{\xi}} \left(\left(\frac{np}{N_u} \right)^{-\hat{\xi}} - 1 \right), \quad (2)$$

where u is the tail cutoff, N_u exceedances, and p exceedance probability.

III. DATASET AND METHODOLOGY

A. Dataset

TABLE I: Elliptic Dataset Statistics

Property	Value	Note
Transactions	203 769	total nodes
Directed edges	234 355	avg. degree 2.30
Illicit (label 1)	4 545	2 % of total
Licit (label 2)	42 019	21 % of total
Unlabeled	157 205	77 % of total
Time steps	49	steps 44–49 unlabeled
Local features	93	1f_1–1f_93
Aggregated feat.	71	af_1–af_71

The Elliptic dataset was published by Elliptic in collaboration with MIT. Feature semantics are not published; columns carry only numerical indices.

TABLE II: Temporal Data Split (by time step index, 1–49)

Partition	Labeled rows	Illicit %
Train (steps 1–34)	29 894	11.6 %
Validation (steps 30–34)	subset of train	—
Test (steps 35–49)	16 670	6.5 %
Unsup. train (steps 1–34)	136 265 all rows	—
Licit-only (steps 1–34)	26 432	0 %

B. Temporal Split and Evaluation Protocol

A strict temporal split is mandatory: Bitcoin transactions have causal ordering, and random shuffling inflates all metrics by allowing interpolation between past and future behavior.

The 44 % relative drop in illicit rate confirms genuine concept drift. **Primary metric**: F1 on the illicit class. Secondary: PR-AUC (threshold-independent, informative under imbalance) and ROC-AUC.

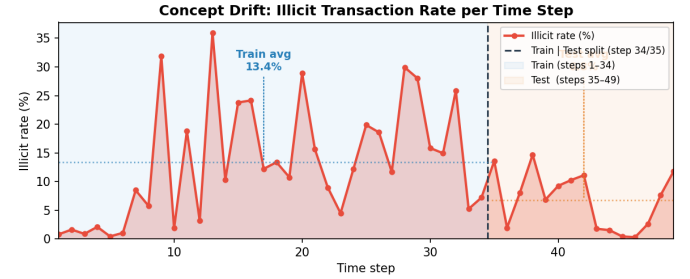


Fig. 1: Illicit transaction rate per time step (labeled nodes only). The rate falls from roughly 11.6 % in the training window (steps 1–34) to 6.5 % in the test window (steps 35–49), a 44 % relative decline confirming genuine concept drift. Random shuffling would mask this shift and inflate all metrics.

C. Per-Paradigm Training Setup

Each paradigm uses different training data and threshold strategy, as summarised in Table III. *Note on supervision*: models in the IsoForest/LOF and OCSVM/AE/VAE rows use no labels for *training*, but their classification thresholds are selected by optimizing F1 on a labeled subset. They are therefore better described as *semi-supervised at threshold-selection stage* rather than fully unsupervised; truly label-free thresholds (EVT) are evaluated separately.

TABLE III: Per-Paradigm Training and Threshold Strategy

Paradigm	Training data	Threshold
IsoForest / LOF	136 265 all train rows	F1-opt, labeled train
OCSVM / AE / VAE	26 432 licit-only rows	F1-opt, val steps 30–34
Supervised	29 894 labeled train	0.5 (default)
GNN	Full graph, train masks	0.5 (default)
Ensemble	Labeled train rows	Grid search, val steps
Semi-sup.	Train + pseudo-labeled	0.5 (default)

D. Model Families

1) *Classical Unsupervised*: Isolation Forest [12] (contamination=0.1); LOF (transductive, applied to test set directly); One-Class SVM (RBF, $\nu=0.05$, licit-only train). Also evaluated: IsoForest on 7 structural topology features, and IsoForest/OCSVM on the top-50 spectral eigenvectors of $\tilde{A} = D^{-1/2}AD^{-1/2}$.

2) *Supervised Tree Models*: All use `class_weight='balanced'` or `scale_pos_weight`: Logistic Regression (L2), Random Forest (100 trees), Gradient Boosting (Optuna-tuned: $n=245$, $\text{depth}=8$, $\text{lr}=0.121$, $\text{sub}=0.918$), LightGBM (500 trees, $\text{max_depth}=8$), RF + graph degree/PageRank features, GBM + 7 structural topology features (degree, clustering, ego-density, OddBall), GBM + 330 temporal rolling features (5-step window), and GBM retrained on pseudo-labeled unlabeled nodes.

3) *Neural Anomaly Detection*: All trained on licit-only data; anomaly score = $-\text{MSE}$ (direction auto-detected on validation via AUC comparison): AE (166 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$... $\rightarrow$ 166), DenoisingAE (latent=8, $\sigma_{\text{noise}}=0.1$), VAE and β -VAE grid search ($\beta \in \{0.01, 1, 2, 4, 8\}$) [7], [11], LSTM-AE (per-timestep mean features, $T=10$ sliding windows), and DOMINANT (2-layer GCN encoder, joint attribute+structure reconstruction) [6].

4) *GNN (Supervised)*: Full PyG graph; edges made undirected; inverse-frequency `CrossEntropyLoss`. GraphSAGE mean aggregation at layer k :

$$h_v^{(k)} = \sigma \left(W^{(k)} \cdot \text{MEAN}(\{h_u^{(k-1)}\} \cup \{h_u^{(k-1)} : u \in \mathcal{N}(v)\}) \right). \quad (3)$$

Variants: baseline (2L, $h=128$, 200ep), GraphSAGEv2 (3L, $h=256$, LayerNorm, self-loops, 250ep), Optuna-tuned ($\text{lr}=0.0038$, $\text{dropout}=0.45$, 274ep), SAGEv2 + pseudo-labels (130 655 nodes), GAT (2L, 4 heads), and GATv2 (3L, 2 heads, residual, LayerNorm).

5) *Ensemble Methods*: Soft-vote ensemble of RF+LGB+GAT (threshold on train, Round 1) and GBM+LGBM+SAGEv2 (threshold grid-searched on validation). Four weight schemes tested; equal weights (1/3 each) won.

E. EVT Thresholding

GPD fitted to the upper tail of reconstruction scores (tail quantile $q=0.90$, exceedance probability $p=0.065$) per Eq. 2. Evaluated on all AE-based models.

F. Implementation

NVIDIA RTX 5070 Ti (12 GB VRAM, CUDA 12.8); Python 3.12, PyTorch 2.11, PyG 2.8, scikit-learn 1.9, LightGBM 4.6, SHAP 0.52, Optuna 4.9, NetworkX 3.6. `StandardScaler` fitted on train only; random seed 42 throughout.

IV. EXPERIMENTS

Experiments follow a logical progression: (1) establish unsupervised floor (RQ1); (2) quantify supervised ceiling and tuning gains; (3) diagnose neural reconstruction ceiling; (4) systematic GNN ablation (RQ3); (5) course-derived advanced

methods; (6) cross-paradigm explainability (RQ4). All models are evaluated on the same 16 670 labeled test rows with no test-set leakage.

A. Unsupervised Baselines

TABLE IV: Classical Unsupervised Anomaly Detection

Model	F1	ROC-AUC	PR-AUC
Isolation Forest	0.021	0.172	0.036
One-Class SVM	0.023	0.235	0.039
LOF	0.076	0.506	0.066
IsoForest (structural)	0.094	0.458	0.056
IsoForest (spectral K=50)	0.056	0.407	0.052
OCSVM (spectral K=50)	0.061	0.391	0.054

All detectors score below $F1 = 0.10$. LOF’s AUC of 0.506 is random. Spectral embedding fails because the Fiedler value ≈ 2.0 (near bipartite maximum) indicates no sparse cut separating illicit nodes—money laundering chains are embedded within the spectral normal subspace.

Answer to RQ1: illicit transactions do *not* form anomalous clusters in feature space. Automated laundering produces stereotyped, regular transactions indistinguishable from high-frequency legitimate activity by density-based methods.

B. Supervised Models

TABLE V: Supervised Model Results (sorted by F1)

Model	F1	ROC-AUC	PR-AUC
Logistic Regression	0.302	0.881	0.291
GBM (baseline)	0.766	0.914	0.785
RandomForest (baseline)	0.801	0.935	0.794
RF + Graph Features	0.807	0.942	0.800
LightGBM	0.817	0.930	0.804
GBM + Temporal Rolling	0.818	0.930	0.802
GBM (Optuna tuned)	0.824	0.920	—
GBM + PseudoLabels	0.824	0.924	0.804
GBM + Structural	0.827	0.924	0.803

Logistic Regression achieves recall = 0.87 but precision = 0.18 ($F1 = 0.302$)—unusable without heavy post-filtering. The jump from LR to RF ($\Delta = +0.499$) reflects non-linear feature interactions that linear models cannot capture. Adding graph-topology features to RF gives only $\Delta = +0.006$ because the 71 aggregated features already encode neighborhood information in tabular form.

GBM Optuna reaches $F1 = 0.824$ ($\Delta = +0.058$ over baseline): the learning-rate–depth–subsample interaction is non-obvious and benefits from systematic search. RF Optuna ($\Delta = -0.004$) confirms RF is near its ceiling.

GBM with 7 structural topology features reaches **$F1 = 0.8265$** , the overall best result ($\Delta = +0.002$ over Optuna GBM): topology information is not fully subsumed by the 71 aggregated features.

GBM with 330 temporal rolling features (5-step window) improves AUC (0.920 $\rightarrow$ 0.930) but reduces F1 (0.824 $\rightarrow$

0.818): global temporal context adds noise to per-transaction discrimination.

Semi-supervised GBM with pseudo-labels reaches $F1 = 0.8235$ ($\Delta = -0.0006$): GBM has already learned the optimal boundary from 29 894 labeled rows.

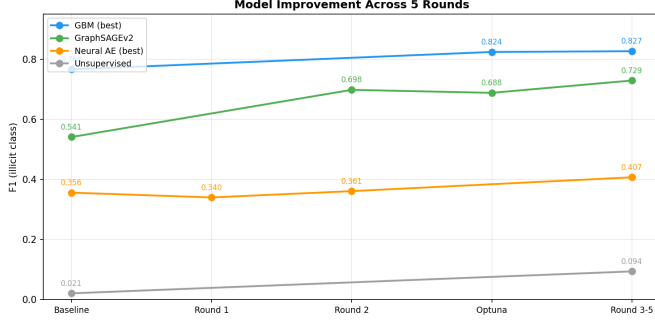


Fig. 2: Improvement across five rounds for key model families. “GBM (best)” tracks the highest-scoring GBM variant at each stage; “Neural AE (best)” tracks the highest-scoring autoencoder variant. Both labels include “(best)” to indicate they represent the family ceiling, not a single fixed model. GBM peaks at $F1 = 0.827$; GraphSAGEv2 reaches 0.729; the autoencoder family peaks at 0.407 (β -VAE, $\beta=0.01$).

C. Neural Anomaly Detection

TABLE VI: Neural Anomaly Detection Results (sorted by F1)

Model	F1	ROC-AUC	PR-AUC
LSTM-AE	0.122	0.446	0.064
VAEv2	0.256	0.732	0.161
β -VAE ($\beta=2.0$)	0.280	0.746	0.211
β -VAE ($\beta=4.0$)	0.313	0.754	0.287
VAE ($\beta=1.0$)	0.340	0.778	0.310
β -VAE ($\beta=8.0$)	0.350	0.763	0.299
AE (baseline)	0.356	0.781	0.336
DenosingAE	0.361	0.792	0.366
β -VAE ($\beta=1.0$)	0.372	0.776	0.338
β -VAE ($\beta=0.01$)	0.407	0.783	0.365

1) *Score Inversion*: The central finding is **score inversion**: illicit transactions reconstruct with *lower* MSE than licit ones, so the anomaly score must be $-MSE$. Automated laundering produces stereotyped, templated patterns that are easier to reconstruct than diverse legitimate activity. Direction is auto-detected on validation by comparing $AUC(y, MSE)$ vs. $AUC(y, -MSE)$.

2) *β -VAE Analysis*: At $\beta=0.01$ (reconstruction-dominated), $F1 = 0.407$ —the best unsupervised non-graph result. As β increases above 1, the KL term compresses licit and illicit representations together, weakening the anomaly signal. The F1 plateau at ≈ 0.36 – 0.41 reflects the hard ceiling of unsupervised reconstruction-based detection on this dataset.

3) *EVT Thresholding*: EVT thresholding (Eq. 2) collapses on DOMINANT and LSTM-AE because score distributions are concentrated near zero with thin tails—the GPD assigns an

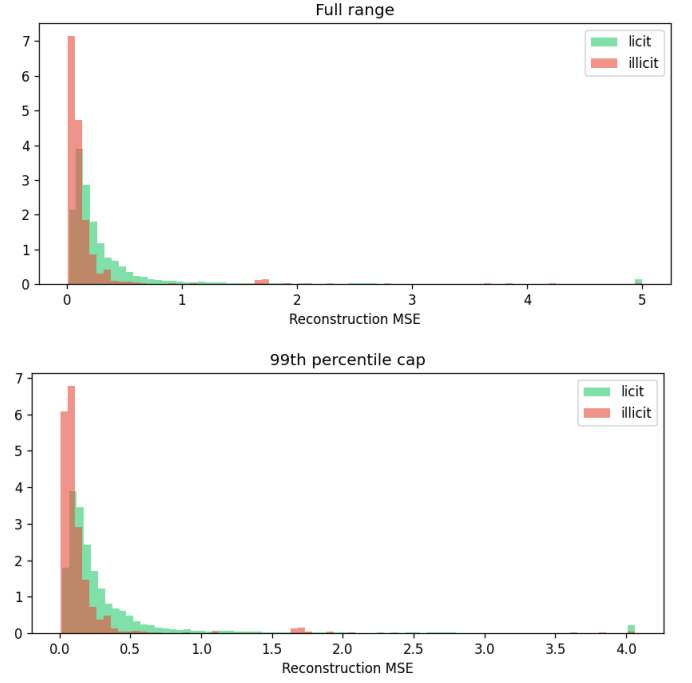


Fig. 3: Autoencoder reconstruction error distributions: histogram (top) and KDE (bottom) for licit (blue) and illicit (orange). Illicit transactions achieve lower MSE, confirming score inversion. Anomaly score is therefore $-MSE$.

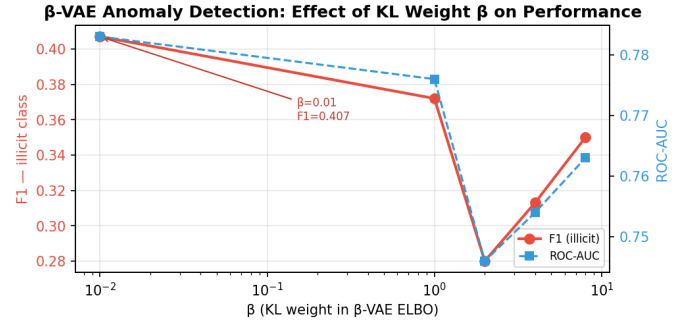


Fig. 4: Effect of KL weight β on β -VAE anomaly detection performance ($F1$ on illicit class and ROC-AUC, log- β axis). Smaller β keeps the objective reconstruction-dominated, preserving the score-inversion signal. $F1$ peaks at $\beta=0.01$ and degrades monotonically above $\beta=1$ as the KL term conflates licit and illicit latent representations.

excessively high threshold, recalling almost no positives. EVT is effective only when reconstruction errors follow genuine heavy-tailed distributions.

4) *LSTM-AE*: The LSTM-AE aggregates features per time step (49×165), trains on 25 sliding windows ($T=10$), and produces a time-step-level anomaly score. ROC-AUC = 0.446 (< 0.5 , inverted) confirms the approach cannot discriminate individual transactions—illicit and licit rows within a step are averaged together. The most anomalous steps detected (40, 46, 39, 43, 36) correspond to high concept-drift periods, not

individual illicit transactions. Correct use: temporal monitoring, not per-transaction scoring.

D. DOMINANT: Graph Autoencoder

DOMINANT [6] jointly reconstructs node attributes and graph structure via a 2-layer GCN encoder (2 layers avoids over-smoothing on this sparse graph). Joint loss:

$$\mathcal{L} = \alpha \|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 + (1 - \alpha) \text{BCE}(\mathbf{A}, \hat{\mathbf{A}}), \quad (4)$$

with $\alpha=0.5$. Result: **F1 = 0.212**, **AUC = 0.669**—the best fully unsupervised method, exceeding all classical detectors by $\Delta > 0.13$. An F1-optimal threshold is required (EVT collapses).

E. GNN Ablation Study

A GNN exploits *guilt by association*: money laundering chains link illicit wallets through intermediate transactions. Ten controlled experiments each isolate one design choice (Table VII).

TABLE VII: GNN Ablation — 10 Controlled Experiments

Exp	Arch	Configuration	F1	AUC	AP
1	SAGE	2L, h=64, 200 ep	0.528	0.887	0.519
2	SAGE	2L, h=128, 200 ep	0.577	0.896	0.603
3	SAGE	2L, h=256, 200 ep	0.580	0.895	0.595
4	SAGE	2L, h=128, 300 ep	0.597	0.897	0.607
5	SAGE	3L, h=128, 200 ep	0.615	0.899	0.611
6	SAGEv2	3L, h=256, LN+SL, 250 ep	0.683	0.906	0.704
7	SAGE	3L, h=256, lr=0.0038, 274 ep (Opt.)	0.690	0.901	0.706
8	GAT	2L, 4 heads, lr=5e-4, 200 ep	0.307	0.824	0.366
9	GAT	2L, 2 heads, lr=1e-3, 200 ep	0.341	0.864	0.419
10	GATv2	3L, 2 heads, residual+LN, 300 ep	0.382	0.887	0.547

1) Ablation Findings:

- **Width** (Exps 1–3, 64→256): +0.052 F1 total with diminishing returns; saturates beyond $h=128$.
- **Training duration** (Exps 2 vs. 4, +100 ep): +0.020 F1; cheapest lever before changing architecture.
- **Depth** (Exps 2 vs. 5, 2L→3L): +0.038 F1; 3-hop receptive field captures longer laundering chains.
- **LayerNorm + self-loops** (Exps 5 vs. 6): +0.068 F1, the largest single gain. LN prevents gradient instability at 3 layers; self-loops preserve node self-features during mean aggregation.
- **Optuna** (Exps 6 vs. 7, lr=0.0038): +0.007 F1; architecture choice dominates over tuning.
- **GAT family** (Exps 8–10): best GAT F1 = 0.382 vs. best SAGE F1 = 0.690 (gap = 0.308). On a sparse graph (avg. degree 2.3), attention weights over too few neighbors are noisy; mean aggregation is more robust.

Answer to RQ3: LayerNorm+self-loops > depth > width > tuning. In this configuration and feature setting, vanilla GAT and GATv2 are counterproductive on a sparse graph (avg. degree 2.3); recent work with Chebyshev convolutions and deeper GATv2 + skip connections has narrowed the gap on Elliptic under different feature subsets, so the finding should not be generalized beyond mean-aggregation architectures at this scale.



Fig. 5: GNN ablation across 10 experiments. Top: F1 progression showing the GraphSAGE improvement trajectory (Exps 1–7) and the GAT family ceiling (Exps 8–10). Bottom: corresponding AUC/AP values confirming the same ranking.

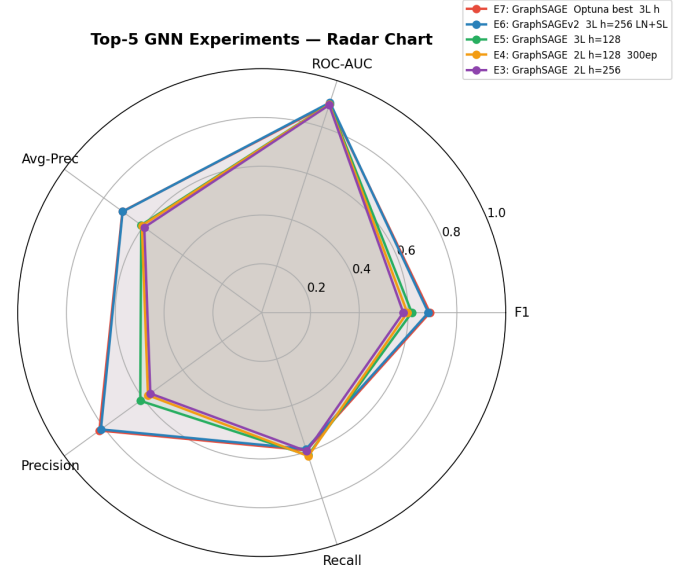


Fig. 6: Radar chart of the top-5 GNN experiments across five metrics: F1, ROC-AUC, Average Precision, Precision, and Recall. GraphSAGEv2 (Exp 6) and the Optuna-tuned variant (Exp 7) dominate across all axes, while GAT family experiments show systematically lower coverage, confirming mean aggregation outperforms attention on this sparse graph.

TABLE VIII: Optuna TPE Tuning Results (40 trials each)

Model	Base F1	Tuned F1	Δ
GBM	0.766	0.824	+0.058
GraphSAGE	0.541	0.688	+0.147
RF	0.801	0.797	−0.004

F. Hyperparameter Tuning (Optuna)

Best parameters: GBM ($n=245$, $\text{depth}=8$, $\text{lr}=0.121$, $\text{sub}=0.918$); GraphSAGE ($\text{hidden}=256$, $\text{dropout}=0.45$, $\text{lr}=0.0038$, 274 ep); RF is near ceiling (-0.004 within search variance). GBM profits from depth-lr interaction; GraphSAGE gains the most absolutely because default $\text{lr}=0.01$ is too aggressive for this sparse graph.

G. Ensemble and Calibration

TABLE IX: Ensemble and Calibration Results

Model	F1	ROC-AUC	PR-AUC
Ensemble (RF+LGB+GAT)	0.742	0.922	0.789
Ensemble (GBM+LGBM+SAGEv2)	0.821	0.924	0.802
Weighted Ensemble (1/3 each)	0.821	0.924	0.802
GBM + Platt calibration	0.801	0.920	0.800
GBM + Isotonic calibration	0.734	0.875	0.768

The GBM+LGBM+SAGEv2 ensemble ($F1 = 0.821$) narrows the gap to GBM solo (0.824) but does not surpass it. Testing four weight schemes shows equal weights win—downweighting SAGEv2 removes useful graph signal. GBM is already well-calibrated (Brier = 0.0196); Platt scaling makes probabilities more conservative (-0.023 F1); isotonic regression overfits validation and distorts ordering (AUC drops $0.920 \rightarrow 0.875$).

H. Full 48-Experiment Leaderboard

Table X reports the top 25 of 48 experiments. Fig. 7 shows the top-15 by F1. Fig. 8 shows Precision–Recall curves for key supervised models, confirming that high AP (≈ 0.80) is achievable across the recall range.

TABLE X: Top-25 of 48 Experiments (by F1 on illicit class)

#	Model	Cat.	F1	AUC	AP
1	GBM + Structural	Sup.	0.827	0.924	0.803
2	GBM (Optuna)	Sup.	0.824	0.920	—
3	GBM + PseudoLabels	Semi	0.824	0.924	0.804
4	Ensemble GBM+LGBM+SAGEv2	Ens.	0.821	0.924	0.802
5	Ensemble equal 1/3	Ens.	0.821	0.924	0.802
6	GBM + Temporal Roll.	Sup.	0.818	0.930	0.802
7	LightGBM	Sup.	0.817	0.930	0.804
8	RF + Graph Features	Sup.	0.807	0.942	0.800
9	RF (baseline)	Sup.	0.801	0.935	0.794
10	GBM + Platt	Sup.	0.801	0.920	0.800
11	RF (Optuna)	Sup.	0.797	0.929	—
12	GBM (baseline)	Sup.	0.766	0.914	0.785
13	Ensemble RF+LGB+GAT	Ens.	0.742	0.922	0.789
14	GBM + Isotonic	Sup.	0.734	0.875	0.768
15	SAGEv2 + PseudoLabels	Semi-GNN	0.729	0.890	0.752
16	GraphSAGEv2	GNN	0.698	0.913	0.720
17	GNN Exp 7 (Optuna)	GNN	0.690	0.901	0.706
18	GraphSAGE (Optuna)	GNN	0.688	—	—
19	GNN Exp 6 (3L LN+SL)	GNN	0.683	0.906	0.704
20	GNN Exp 5 (3L h=128)	GNN	0.615	0.899	0.611
21	GraphSAGE (baseline)	GNN	0.541	0.888	0.549
22	β -VAE ($\beta=0.01$)	Neural	0.407	0.783	0.365
23	DenoisingAE	Neural	0.361	0.792	0.366
24	DOMINANT	Graph-AE	0.212	0.669	0.123
25	LSTM-AE	Neural	0.122	0.446	0.064

Five performance tiers:

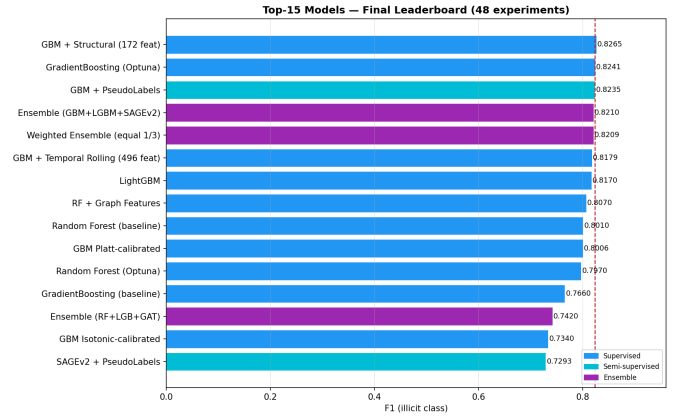


Fig. 7: Top-15 models by F1 on illicit class. The dashed line marks the GBM Optuna baseline (0.824) for reference. All top-15 entries are supervised or ensemble-based.

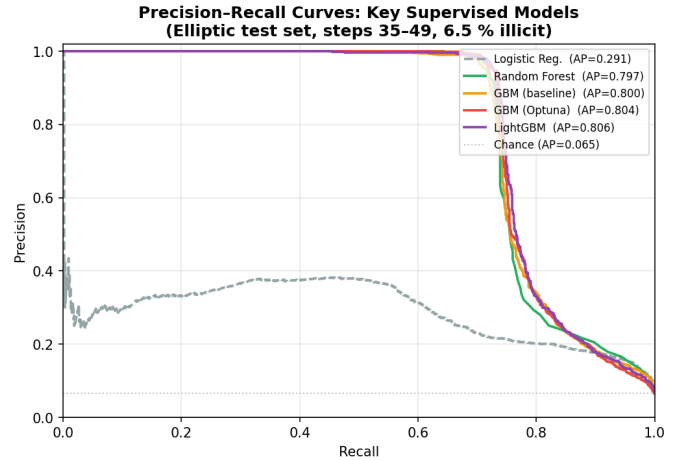


Fig. 8: Precision–Recall curves for key supervised models on the test set (steps 35–49, 6.5 % illicit). LightGBM and GBM (Optuna) achieve $AP \approx 0.80$, maintaining high precision across the recall range. Logistic Regression collapses to low precision at moderate recall ($AP = 0.291$). The dotted line marks the chance baseline ($AP = 0.065$). PR-AUC is threshold-independent and more informative than ROC-AUC under severe class imbalance.

- 1) **Supervised/Ensemble** (F1 0.74–0.83): tree models dominate due to 71 aggregated features pre-encoding graph signal.
- 2) **GNN supervised** (F1 0.54–0.73): GraphSAGEv2 recovers 84.5 % of the supervised ceiling using only topology.
- 3) **Neural AE** (F1 0.12–0.41): bounded by reconstruction ceiling; best from β -VAE ($\beta=0.01$).
- 4) **Graph AE unsupervised** (DOMINANT, F1 = 0.212): best unsupervised *graph* method (β -VAE is higher overall).
- 5) **Classical unsupervised** (F1 < 0.10): density methods fail.

Answer to RQ2: graph structure alone recovers 84.5 % of the supervised ceiling. GBM and GraphSAGEv2 exploit different, complementary channels (tabular features vs. topology).

I. Explainability

TABLE XI: Cross-Model Feature Agreement (top-20 lists)

Agreement	Features
$\text{RF} \cap \text{GBM}$ (7)	1f_18, 1f_47, 1f_53, 1f_59, 1f_76, 1f_90, af_70
$\text{RF} \cap \text{GBM} \cap \text{GNN}$ (3)	1f_53, 1f_90, af_70

SHAP TreeExplainer was applied to GBM (Optuna, the best tabular model, $F1 = 0.824$) and RF; gradient attribution was computed for GraphSAGE by backpropagating the illicit-class logit to the input and averaging over illicit test nodes. Fig. 9 shows the GBM beeswarm as the primary explainability view.

Three features—1f_53, 1f_90 (local), and af_70 (aggregated neighborhood)—rank top-20 in every model family. Their cross-paradigm consensus validates them as genuine signals, not model-specific artifacts. The presence of the aggregated feature af_70 in GNN attribution confirms that graph structure encodes real anomaly signal.

Answer to RQ4: robust cross-paradigm signals exist spanning both local transaction statistics and neighborhood aggregates. Identifying their semantics via Elliptic collaboration is the primary open question.

V. DISCUSSION

RQ1 — Density failure. All classical detectors score below $F1 = 0.076$. Illicit transactions are stereotyped, not rare. This directly contradicts the assumption of classical anomaly detection—that anomalies are low-density outliers. On the Bitcoin blockchain, money-laundering patterns (chain peeling, mixing) are individually indistinguishable from legitimate high-frequency activity.

RQ2 — Graph vs. tabular. GraphSAGEv2 ($F1 = 0.698$) recovers 84.5 % of GBM ($F1 = 0.824$). The remaining 15.5 % gap reflects the richer signal encoded in the 71 pre-computed aggregated features available to tree models. A stacking ensemble of GBM + GraphSAGEv2 is the natural next step.

RQ3 — GNN design. LayerNorm+self-loops (+0.068) > depth (+0.038) > width (+0.052 total) > learning-rate tuning (+0.007). Mean aggregation dominates attention on sparse graphs.

RQ4 — Robust features. 1f_53, 1f_90, af_70 are top-20 across all model families. The aggregated feature af_70 in GNN attribution confirms that graph neighborhood signal is captured independently by tabular proxies in tree models.

Autoencoder paradox. Illicit transactions are *more* regular than licit ones. Score inversion (low MSE = anomaly) must be validated on labeled data before deploying any reconstruction-based AML system.

β -VAE. Lower β preserves reconstruction dominance; $\beta \geq 2$ compresses licit and illicit representations together. The ceiling

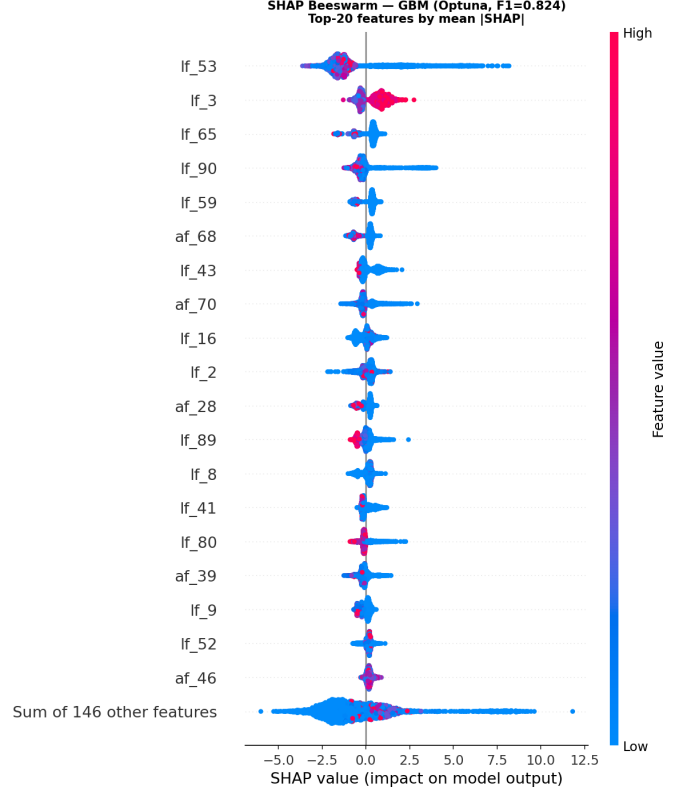


Fig. 9: SHAP beeswarm for GBM (Optuna, $F1 = 0.824$), the best tabular model. Each dot is one test-set transaction (sample of 3 000); color encodes feature value (red=high, blue=low). Positive SHAP pushes toward illicit; negative toward licit. Features 1f_53, 1f_90, and af_70 rank in the top band and appear consistently across RF, GBM, and GraphSAGE gradient attribution.

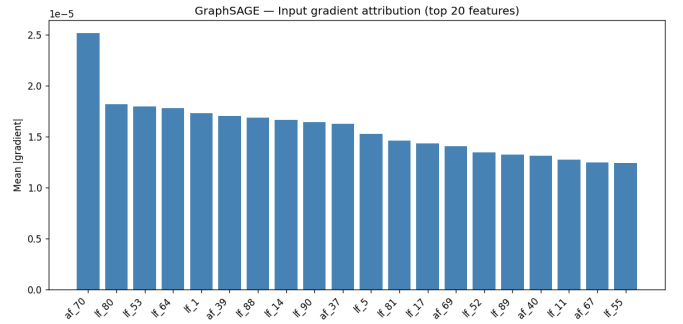


Fig. 10: GraphSAGE gradient attribution: mean absolute gradient per feature, averaged over illicit test nodes. Features 1f_53, 1f_90, and af_70 are prominent despite the entirely different learning mechanism, confirming the cross-paradigm consensus.

at $F1 \approx 0.41$ is the fundamental limit of unsupervised reconstruction-based detection here.

Semi-supervised pseudo-labels. GBM gains $\Delta = -0.0006$: supervised boundary is already saturated. GraphSAGEv2 gains $+0.031$: extra pseudo-labeled nodes populate graph neighborhoods during message passing.

Concept drift. Illicit rate drops from 11.6% to 6.5% over 15 steps. Production systems need temporal threshold recalibration; practical strategies include sliding-window F1 re-optimization on the most recent labeled window, Bayesian prior-shift correction, and adaptive class-weight schedules tied to rolling illicit-rate estimates.

Limitations. (1) Feature semantics unknown; robust features cannot be interpreted without Elliptic collaboration. (2) GAT Optuna tuning incomplete; recent Elliptic work with Chebyshev convolutions and skip-connected GATv2 reports higher GNN F1 under different feature subsets, so our GNN results represent a strong but not exhaustive upper bound. (3) GBM + GraphSAGEv2 stacking unexplored. (4) DOMINANT with more epochs and α tuning may gain further. (5) Per-transaction LSTM-AE (vs. timestep aggregate) not evaluated. (6) This work positions itself as a comprehensive baseline benchmark across paradigms; direct comparison to reported Elliptic state-of-the-art (e.g., Chebyshev-GCN, N2V-GCN) is left for future work.

VI. CONCLUSION

We benchmarked 48 model variants on the Elliptic dataset across five improvement rounds. Six actionable findings:

- 1) **Temporal split is mandatory.** Random shuffling leaks future patterns; the 11.6% $\rightarrow$ 6.5% illicit-rate drift is real.
- 2) **Supervised trees dominate.** GBM + structural topology reaches $F1 = 0.827$; 71 aggregated features encode enough graph signal to outperform unsupervised GNNs.
- 3) **GNNs add complementary structure.** GraphSAGEv2 reaches $F1 = 0.698$ using only graph topology. 3-layer mean aggregation with LayerNorm and self-loops is optimal; GAT consistently underperforms on sparse graphs.
- 4) **Reconstruction detection is viable but inverted.** β -VAE ($\beta=0.01$, $F1 = 0.407$) is the best unsupervised non-graph method. Score direction must be validated before deployment.
- 5) **DOMINANT is the best unsupervised graph-autoencoder method.** $F1 = 0.212$, $AUC = 0.669$, exceeding all classical density detectors by >0.13 $F1$ (β -VAE, tabular, reaches $F1 = 0.407$).
- 6) **Three features are cross-paradigm robust.** `lf_53`, `lf_90`, `af_70` rank top-20 in RF, GBM (SHAP), and GraphSAGE (gradient attribution).

Future work: (1) GBM + GraphSAGEv2 stacking ensemble; (2) per-transaction LSTM-AE; (3) EvolveGCN [2] for temporal drift; (4) online threshold recalibration; (5) GNNExplainer for node-level attribution; (6) Elliptic collaboration to identify feature semantics.

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